

AUTOMATED NETWORK REQUEST MANAGEMENT

ServiceNow Custom Application

Complete Project Documentation

Platform: ServiceNow

Project: Automated Network Request Management

Scope: Service Catalog, Approvals and Flow Designer Automation

Document: Implementation, Catalog, Automation and Testing Report

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1. Project Overview

The Automated Network Request Management project is a ServiceNow solution that lets employees and network teams submit, route, approve, and fulfil network-related service requests through a structured catalog record and Service Catalog experience. The implementation combines a custom network request table, catalog-driven form design, conditional UI Policies, an approval relationship, and Flow Designer automation.

- Requesters submit network requests (connections, access, devices) through a guided catalog form.
- Requests are stored in a custom network request table alongside standard platform tables.
- Conditional UI Policies reveal additional fields based on selections (e.g. connection type, device type).
- An Approval Request relationship links approval records (sysapprovals) to each network request.
- Flow Designer automates record creation, approval routing, email notification, and status updates.
- List views give approvers and network teams centralized visibility into requests, tasks, and statuses.

2. Problem Statement and Objectives

Network teams need a simple, consistent method to receive and process network-related service requests. A structured ServiceNow solution reduces unorganized, manual requests and gives network teams a consistent record containing requester and request information.

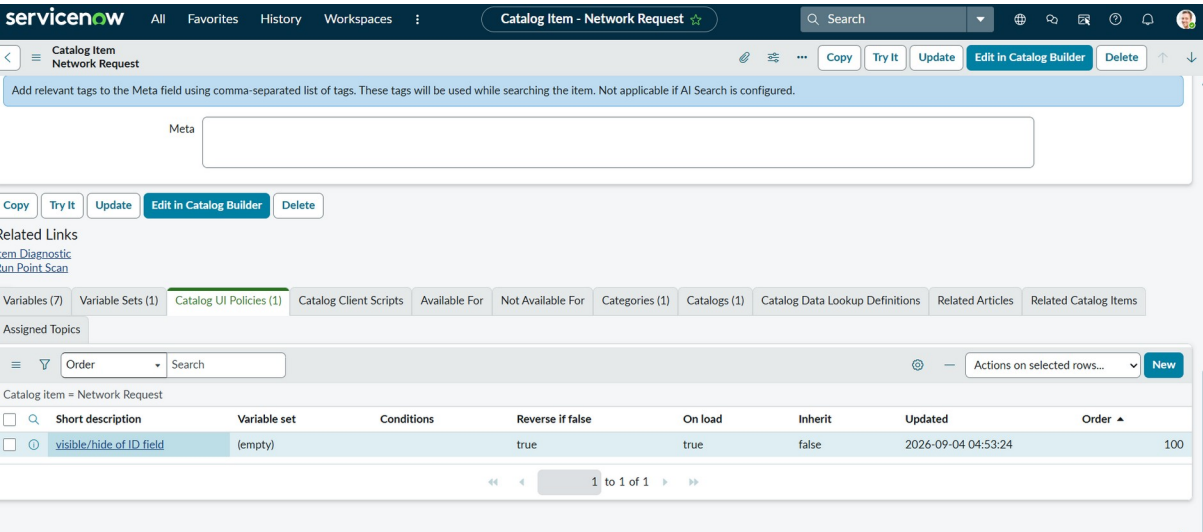
Objectives

- Streamline and automate the end-to-end lifecycle of network-related service requests.
- Reduce manual effort and human error in request intake and routing.
- Accelerate request fulfilment times.
- Enforce standardized workflows and approval processes.
- Improve visibility and tracking through a centralized platform.
- Enhance end-user experience with faster, more reliable service delivery.
- Ensure compliance with IT and security policies.

3. Application and Catalog Item Creation

The functional scope covers catalog creation, form design, approval routing, Flow Designer automation, and email notification. A Network Request catalog item is configured within the Service Catalog, mapped to the custom network request table, with stakeholder mapping and an execution roadmap defined during the design phase.

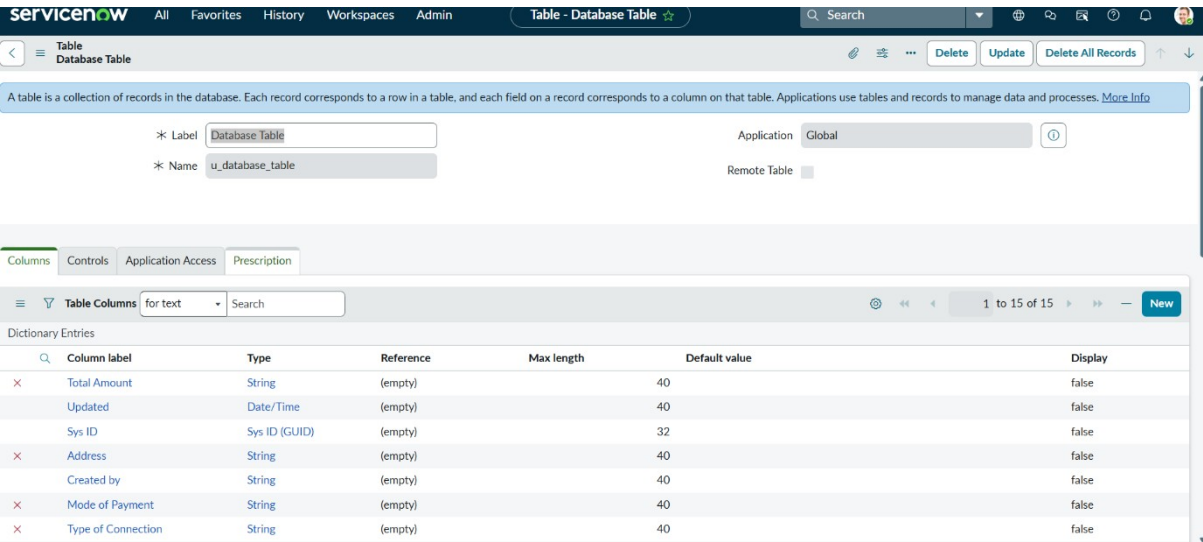
The screenshot shows the ServiceNow 'Catalog Item - Network Request' configuration page. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The main header displays 'Catalog Item - Network Request' with a search icon and a dropdown menu. Below the header, there are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue informational box states: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. • Enter a Name and Short description to display for the item. • Enter a Price, approvals, variables, and other information as needed.' The form fields include: 'Name' (Network Request), 'Application' (Global), 'Catalogs' (Service Catalog), 'Category' (Network Standard Changes), 'State' (None), 'Checked out' (None), 'Owner' (System Administrator), 'Active' (checked), and 'Fulfillment automation level' (Unspecified). The bottom section, titled 'Item Details', contains a 'Short description' (Network Services Request) and a 'Description' field with a rich text editor toolbar.



4. Custom Table and Data Architecture

The Automated Network Request Management solution is built on a structured data model that combines standard platform tables with a custom network request table for clarity, scalability, and control. Network requests are captured through catalog-driven records and stored with well-defined fields such as request type, access level, business justification, and related network configuration items.

Forms are designed to dynamically display only relevant information using UI logic, while mandatory fields enforce data accuracy and compliance. List views provide operational visibility for approvers and network teams, enabling efficient tracking of requests, tasks, and statuses. This architecture ensures secure access, seamless automation, and reliable reporting across the entire request lifecycle.



The screenshot displays two parts of the ServiceNow interface. The top part shows the 'Table - Database Table' configuration page, which lists various fields and their properties. The bottom part shows the 'Database Table - DAT00001010' form, which contains input fields for various attributes and a table for 'Approvals'.

Table - Database Table Configuration:

Field	Type	Value	Length	Script	Index
Type of Connection	String	(empty)	40		false
AssignedTo	Reference	User	32		false
Approval Status	String	(empty)	40		false
Database Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Created	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Requested For	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Mobile Number	String	(empty)	40		false

Database Table - DAT00001010 Form:

Database Number: DAT00001010
 Mobile Number: 9090909090
 Total Amount: 800
 Mode of Payment: card
 Requested For: Sampath
 Type of Connection: existing
 Address: Hyderabad
 Assignment To: ITIL User
 Approval Status: Approved

Approvals Table:

State	Approver	Comments	Approval for	Created
Approved	(empty)		(empty)	2026-09-04 03:06:46

5. Approval Request Relationship

Creation of Approval Request on the Network Database Table

Activity 1: Creation of Related List

- Navigate to System Definition > Relationships.
- Click New to create a new relationship.
- Name: Approval Request.
- Applies to Table: the custom network request (database) table.
- Queries from Table: the sysapprovals table.
- Active: set to True.
- Save the relationship.

The screenshot shows the ServiceNow 'Relationships' list view for 'Approval Request'. The table lists various relationships with columns for Name, Advanced, Apply to, Applies to table, Queries from table, Insert callback, Query from, Query with, and Updated. The first row is selected, showing details for 'Approval Request'.

Name	Advanced	Apply to	Applies to table	Queries from table	Insert callback	Query from	Query with	Updated
Approval Request	false		Database Table [u_database_table]	Approval [sysapproval_approver]			(function refineQuery(current, parent) [...])	2026-09-04 01:03:40
Approvals	false	var gr = new GlideRecord(table_name); an...	Knowledge [kb_knowledge]	Approval [sysapproval_approver]		answer = 'sysapproval_approver';	current.addQuery('source_table', parent...)	2017-09-07 03:00:53
Approvals	false		Business Process [cmdb_ci_business_process]	Approval [sysapproval_approver]			(function refineQuery(current, parent) [...])	2020-09-23 08:47:13
Approvers	false		Publication [sn_publications_publication]	Approval [sysapproval_approver]			(function refineQuery(current, parent) [...])	2016-01-17 16:19:56
Archive Rules	false		Data Management Policy [sys_dm_policy]	Archive Rule [sys_archive]			(function refineQuery(current, parent) [...])	2024-01-18 13:47:37
Article Versions	false		Knowledge [kb_knowledge]	Knowledge Version [kb_version]			(function refineQuery(current, parent) [...])	2017-01-25 02:47:46
Associated Roles	false		Product Family [ua_app_family]	Associated Roles [assoc_roles]			(function refineQuery(current, parent) [...])	2020-04-01 16:36:08

The screenshot shows the ServiceNow 'Relationship - Approval Request' configuration page. It includes fields for Name, Application, Applies to table, and Queries from table. Below these fields is a code editor for the 'Query with' field, containing a JavaScript function to refine the query. The page also has buttons for 'Run Query Diagnostics', 'Update', and 'Delete'.

Name: Approval Request

Application: Global

Applies to table: Database Table [u_database_table]

Queries from table: Approval [sysapproval_approver]

This script refines the query in current that will populate the related list. For more information about it, its parameters and control variables, see the documentation. See also the article about the recommended form of the script.

Query with: ☐ Turn on ECMAScript 2021 (ES12) mode

```
1 (function refineQuery(current, parent) {  
2  
3   // Add your code here, such as current.addQuery(field, value);  
4   current.addQuery('source_table', parent.getTabletName());  
5   current.addQuery('document_id', parent.sys_id);  
6  
7 })(current, parent);
```

Run Query Diagnostics Update Delete

Related Links
Run Point Scan

This relationship surfaces approval records directly on the network request form as a related list, giving requesters and approvers a single place to track approval status.

6. Form Design and Field Configuration

The Network Request form is designed to collect the information required to process a request, separating system-generated information (such as request number) from user-entered or selected information (such as request type, access level, business justification, and related network configuration items).

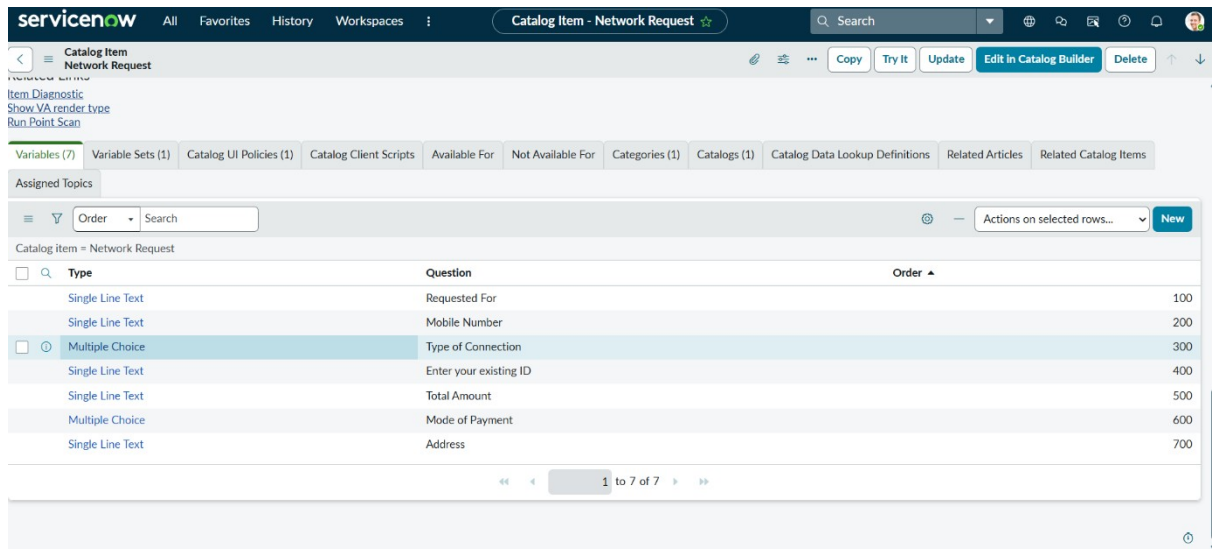
variables

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010006	Software	(empty)	Provide requested service	(empty)	(empty)	200
SCTASK0010005	Field Services	(empty)	Assess or Scope Task	2026-09-10 05:59:45	(empty)	100

7. Catalog Variables Configuration

Variables are configured on the Network Request catalog item to build the form fields presented to the requester.

- Variable types include Single Line Text, Multi Line Text, Reference, and Choice, selected per requirement.
- Order values (100, 200, 300, ...) control the display sequence of fields.
- Question: the label shown to the requester.
- Name: the internal variable name used for scripting.
- Tooltip: guidance shown when the cursor hovers over the field.
- Example text: sample input to guide the requester.
- Mandatory / Read-only: configured as required per field.
- Auto-populate: dependent variables use dot-walking to pull related values automatically.

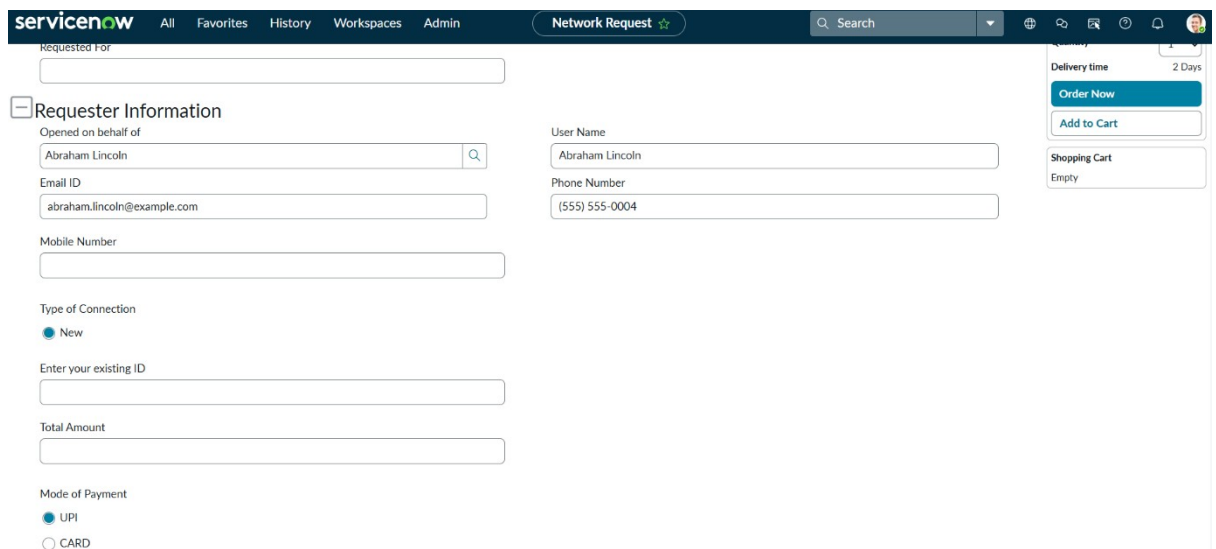


Type	Question	Order
Single Line Text	Requested For	100
Single Line Text	Mobile Number	200
Multiple Choice	Type of Connection	300
Single Line Text	Enter your existing ID	400
Single Line Text	Total Amount	500
Multiple Choice	Mode of Payment	600
Single Line Text	Address	700

8. Navigation Flow

Procedure:

- Log in to the ServiceNow instance.
- Copy the instance domain, e.g. <https://dev190678.service-now.com>.
- Append the /sp prefix to reach the Service Portal, e.g. <https://dev190678.service-now.com/sp>.
- Search for "Network Requests".
- Fill in the required details and click Submit.
- A new request is generated with a request number, and the requester receives a confirmation email.



Requested For

Requester Information

Opened on behalf of

Abraham Lincoln

Email ID

abraham.lincoln@example.com

Mobile Number

Type of Connection

New

Enter your existing ID

Total Amount

Mode of Payment

UPI

CARD

User Name

Abraham Lincoln

Phone Number

(555) 555-0004

Delivery time

2 Days

Order Now

Add to Cart

Shopping Cart

Empty

Order Status

Thank you, your request has been submitted

Order Placed: 2026-09-04 04:34:45
Request Number: [REQ0010001](#) ☆
Estimated Delivery Date of Complete Order: 2026-09-06

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
Network Services Request	2026-09-06	▶ ○ ○ ○ ○		1	
					Total

[Back to Catalog](#) [Continue Shopping](#) [Home](#)

9. UI Policy and Conditional Validation

Catalog UI Policy Configuration

UI Policies display additional fields when certain choices are made, using conditional logic and field dependencies configured within ServiceNow.

Scenario 1: If the user selects "types of Connection" = Existing, the Existing ID field is displayed. Scenario 2: If the user selects "types of devices" = Others, the "Please specify" field is displayed.

Procedure:

- Navigate to the Network Request catalog item.
- Open the Catalog UI Policy related list and click New.
- Set Applies to = Catalog item, and Catalog item = Network Request.
- Provide a short description and the triggering condition (e.g. types of Connection is Existing).
- Save, then configure the UI Policy Action for the target variable and set it Visible = True.
- Update the UI Policy and test it on the catalog form.

Catalog UI Policy - visible/hide of ID field

Requested Item or Catalog Task form: [More Info](#)

Applies to: A Catalog Item Application: Global

* Catalog item: Network Request Active: ☒

* Short description: visible/hide of ID field

When to Apply Script

Catalog UI policy actions are applied only if all the following conditions are met:

1. The catalog UI policy is **Active**
2. The items in the **Conditions** field evaluate to true
3. The field specified in the catalog UI policy is present on the specified catalog item

Catalog Conditions: [Add Filter Condition](#) [Add OR Clause](#)

type_of_connection is Existing AND OR

Applies on a Catalog Item view: ☒ Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form

Applies on Catalog Tasks: ☒

Applies on Requested Items: ☐ On load: ☒ Reverse the effects of the catalog UI policy actions when the Conditions evaluate to false

The screenshot shows the 'Catalog UI Policy' configuration page in ServiceNow. The policy is named 'visible/hide of ID field'. It has three checkboxes for where it applies: 'Applies on a Catalog Item view' (checked), 'Applies on Catalog Tasks' (checked), and 'Applies on Requested Items' (unchecked). There are two blue action boxes: 'Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form' and 'Reverse the effects of the catalog UI policy actions when the Conditions evaluate to false'. The 'On load' checkbox is checked, and 'Reverse if false' is also checked. At the bottom, there is a table of 'Catalog UI Policy Actions'.

Name	Read only	Mandatory	Visible	Order
enter_your_existing_id	Leave alone	Leave alone	True	100

10. Automation — Flow Designer

Flow Designer logic implements: Get Catalog Variables, Create Record, Ask for Approval, Send Email, Update Record — triggered on catalog item submission and supporting dynamic automation.

Activity 1: Creation of Flow

- Navigate to the Flow Designer home page.
- Click New to create a new flow.
- Provide the flow name: Network Request.
- Provide a description of the flow.
- Click Build flow, then add the Get Catalog Variables, Create Record, Ask for Approval, Send Email, and Update Record actions in sequence.

The screenshot shows the 'EXECUTION DETAILS' for a 'Network Request' flow. The flow is currently in a 'Waiting' state. The execution details table shows the following steps:

Step	Action	Type	State	Start time	Duration
1	Get Catalog Variables from Network Request	Core Action	Completed	2026-09-10 06:29:55	20ms
2	Create Record	Core Action	Completed	2026-09-10 06:29:55	6ms
3	Send Email	Core Action	Completed	2026-09-10 06:29:55	28ms
4	Ask For Approval	Core Action	Waiting	2026-09-10 06:29:55	55ms
5	If If Request is Approved	Flow Logic	Not Run		
6	Update Record	Core Action	Not Run		
7	Else	Flow Logic	Not Run		
8	Update Record	Core Action	Not Run		

Workflow Studio | Network Request | Operations | Network Request

EXECUTION DETAILS | **Network Request** | Test Run - Completed | Open flow | Open context record

Show Action Details

		State	Start time	
1	Get Catalog Variables from Network Request	Core Action	Completed	2026-09-10 06:29:55 20ms
2	Create Record	Core Action	Completed	2026-09-10 06:29:55 6ms
3	Send Email		Completed	2026-09-10 06:29:55 28ms
4	Ask For Approval	Core Action	Completed	2026-09-10 06:29:55 144ms
5	If Request is Approved	Flow Logic	Evaluated - True	2026-09-10 06:34:23 8ms
6	Update Record	Core Action	Completed	2026-09-10 06:34:23 8ms
7	Else	Flow Logic	Not Run	
8	Update Record	Core Action	Not Run	

ERROR HANDLER

11. Access Control

Role-based access control restricts who can view, create, and update network request records and approvals, ensuring that only authorized requesters, approvers, and network team members can act on a given request.

State	Approver	Comments	Approval for	Created
Approved	System Administrator		(empty)	2026-09-13 02:30:54
Approved	System Administrator		(empty)	2026-09-10 06:29:55
Approved	System Administrator		(empty)	2026-09-13 00:18:37
Approved	System Administrator		(empty)	2026-09-13 02:04:42
Approved	System Administrator		(empty)	2026-09-13 02:29:30
Approved	System Administrator		(empty)	2026-09-13 03:38:30
Approved	System Administrator		(empty)	2026-09-06 06:19:34
Approved	System Administrator		(empty)	2026-09-10 06:29:48

servicenow | All | Favorites | History | Workspaces | Admin | **Approvals** | Search | Actions on selected rows...

All > Sys ID = NULL...or, Approver = System Administrator

State	Approver	Comments	Approval for	Created
Approved	System Administrator		(empty)	2026-09-13 02:30:54
Approved	System Administrator		(empty)	2026-09-10 06:29:55
Approved	System Administrator		(empty)	2026-09-13 00:18:37
Approved	System Administrator		(empty)	2026-09-13 02:04:42
Approved	System Administrator		(empty)	2026-09-13 02:29:30
Approved	System Administrator		(empty)	2026-09-13 03:38:30
Approved	System Administrator		(empty)	2026-09-06 06:19:34
Approved	System Administrator		(empty)	2026-09-10 06:29:48

Approval
Database Table: RITM0010008

Approver: System Administrator | Approving: Database Table: RITM0010008

State: Approved

Approval Reason:

Comments: Comments

Post

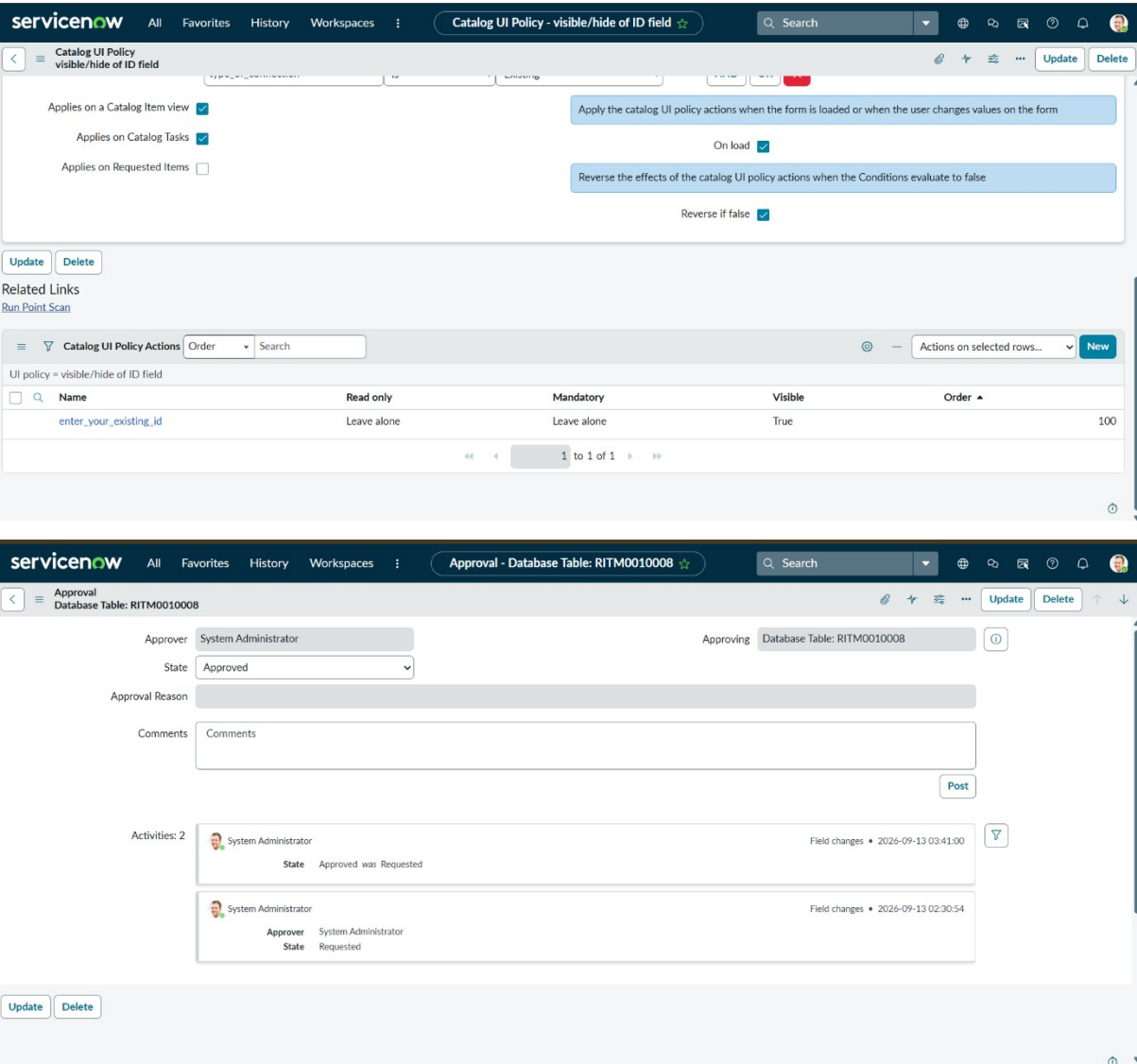
Activities: 2

- System Administrator | Field changes: 2026-09-13 03:41:00 | State: Approved was Requested
- System Administrator | Field changes: 2026-09-13 02:30:54 | Approver: System Administrator, State: Requested

Update | Delete

12. QA Testing

Functional QA testing covered catalog submission, conditional UI Policy behavior, approval routing, automation execution, and record visibility for approvers and network teams.



13. Data Integrity

Mandatory fields, reference fields, and dependent/auto-populated variables enforce data accuracy across the request lifecycle, ensuring that records entering the network request table are complete and consistent.

The screenshot shows a ServiceNow interface for a request record. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header displays 'Request - REQ0010003' with a search bar and action buttons: 'Discuss', 'Follow', 'Update', 'Cancel Request', 'Copy', and 'Delete'. The form fields are organized into two columns. The left column includes 'Number' (REQ0010003), 'Requested for' (System Administrator), 'Location', 'Due date' (2026-09-12 05:59:43), 'Price' (\$0.00), 'Description', 'Short description', 'Special instructions', and 'Comments (Customer visible)'. The right column includes 'Opened' (2026-09-10 05:59:44), 'Opened by' (System Administrator), 'Approval' (Approved), and 'Request state' (Approved). Below the form fields, there is an 'Activities' section showing a single activity by 'System Administrator' and a 'Field changes' section showing a change on '2026-09-10 05:59:44'.

14. Functional Flow



Figure 1: End-to-end functional flow — Requester to notification.

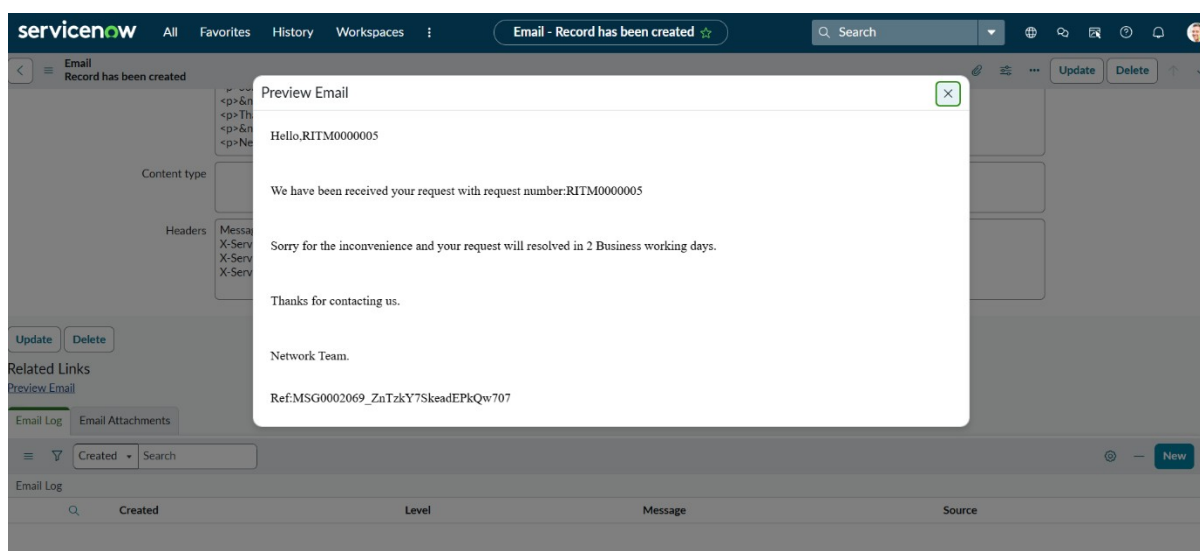
Step	Component	Function
1	Service Catalog	Requester submits the Network Request catalog item.
2	Catalog Variables	Capture request type, access level, justification and related items.
3	UI Policy	Display conditional fields based on connection/device type.
4	Custom Table	Store the network request record.
5	Approval Request Relationship	Link the request to its sysapprovals approval record.
6	Flow Designer	Create record, request approval, send email, update record status.
7	List Views	Give approvers and network teams visibility into requests and status.
8	Notification	Requester receives email confirmation and status updates.

End-to-end architecture: Requester → Service Portal → Network Request catalog item → Custom Table → Approval Request relationship → Flow Designer automation → Status-based tracking.

15. Testing and Validation

ID	Test	Expected / Evidence
TC-01	Submit network request	Request record is created in the custom network request table.
TC-02	Mandatory field validation	Form blocks submission until required fields are completed.
TC-03	Connection type = Existing	Existing ID field becomes visible per UI Policy.
TC-04	Device type = Others	"Please specify" field becomes visible per UI Policy.
TC-05	Approval relationship	Approval Request related list shows the linked sysapprovals record.
TC-06	Flow Designer execution	Get Catalog Variables, Create Record, Ask for Approval, Send Email and Update Record execute successfully on submission.
TC-07	Notification	Requester receives an email confirming submission.
TC-08	List view visibility	Approvers and network teams see the request and its current status.

Emails Created						
All > Created on Today						
☐	Created	Recipients	Subject	Type	Notification type	User ID
☐	2026-09-02 20:59:51	abraham.lincoln@example.com	Employee Onboarding Request - HR Approval Required	send-ready	SMTP	(empty)
☒	2026-09-02 20:59:51	abraham.lincoln@example.com	Employee Onboarding Request Awaiting Your Approval	send-ready	SMTP	(empty)
☐	2026-09-02 01:00:10	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
☐	2026-09-02 01:00:09	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
☐	2026-09-02 20:58:40	abel.tuter@example.com	Employee Onboarding Request Awaiting Your Approval	send-ready	SMTP	(empty)
☐	2026-09-02 10:01:09	admin@example.com	[ITSM Analytics] Daily Data Collection completed with warning	send-ready	SMTP	(empty)
☐	2026-09-02 01:00:10	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
☐	2026-09-02 00:00:09	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)



A recorded walkthrough of the implementation is available at:

https://drive.google.com/file/d/1gDrM3CmFP848MBk_zRy7qVAJv1ECrgzL/view?usp=sharing

16. Conclusion and Future Enhancements

The Automated Network Request Management project demonstrates an end-to-end ServiceNow solution covering catalog-based intake, structured data storage, conditional form logic, approval tracking, and Flow Designer automation.

Key Achievements

- Structured data architecture combining standard and custom network request tables.
- Network Request catalog item with ordered, well-configured variables.
- Conditional UI Policies for connection type and device type scenarios.
- Approval Request relationship linking requests to sysapprovals records.
- Flow Designer automation covering the full request lifecycle.
- Role-based access control and validated data integrity.

Future Enhancements

- SLA-based escalation and automated reassignment for overdue requests.
- Deeper CMDB integration for network configuration items.
- Reporting dashboards for request volume, approval turnaround and fulfilment time.
- Expanded, multi-level approval hierarchies.
- Additional request types and knowledge-article recommendations during submission.

End of Project Documentation
Automated Network Request Management — ServiceNow